

A MR-Mapping System for Guided Self-Correction in Postural Rehabilitation

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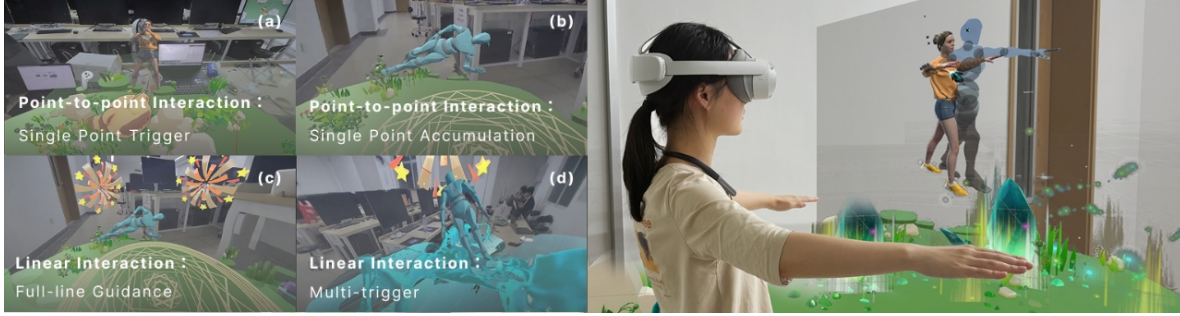


Figure 1: Overview of the Trainer-Anchored MR system for guided self-correction.

ABSTRACT

For large-limb movement correction training scenarios, real-time correction and standard movement observation during the process are important requirements. Using computer vision to achieve real-time recognition of user postures and constructing personalized training feedback streams through dynamic interaction paths are novel channels to enhance the experience. In this study, we propose a Trainer-Anchored MR system for large-limb rehabilitation training, in order to achieve the goal of self-correction of movement learning with low cost and high engagement.

Index Terms: Human-centered computing—Mixed reality—Interaction techniques; Human-centered computing—Human computer interaction (HCI)—Interactive systems and tools

1 INTRODUCTION

With the advancement of mixed reality (MR) technologies, overlaying reference information directly into real-world environments has become a promising approach for guided movement training. Compared to conventional screen-based instructions, MR-based guidance allows users to perceive corrective cues without interrupting full-body motion, which is especially important for large-limb exercises [1]. Prior work has explored first-person MR perspectives [2], embedded reference videos, and trainer-anchored displays [3]. However, many existing approaches rely on manual preprocessing or dedicated motion-capture systems, limiting scalability and practical deployment.

To overcome these limitations, we propose an automated, vision-based MR pipeline that converts online training videos into real-time, trainer-anchored guidance. Our system aligns user and trainer poses for continuous error estimation and further incorporates gaze-driven attention cues from the trainer to generate implicit visual guidance during movement execution.

Our contributions are threefold:

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- We present a low-cost, fully automated vision-based pipeline that transforms arbitrary online training videos into immersive MR guidance without motion capture.
- We propose a trainer-anchored alignment and gaze mapping method that supports real-time self-correction through overlaid motion and attention cues.
- We implement an end-to-end interactive MR prototype integrating perception, guidance, and feedback for functional movement training.

2 SYSTEM IMPLEMENTATION

This system is based on mixed reality, combining structured video processing, vision-based pose recognition and spatial geometric mapping methods to construct a set of self-correcting real-time interactive system for functional movement training, which is composed of three core modules.

Spatial Perception and Pose Alignment Module: This module extracts body keypoints from both the reference video and the user's real-time camera feed using Mediapipe Pose. A composite center point is computed from the shoulders and hips, and a torso-based scale factor is used to normalize body size differences. The user's pose is then aligned to the reference using an affine transformation:

$$\mathbf{p}_i^{u'} = s \cdot (\mathbf{p}_i^u - \mathbf{c}^u) + \mathbf{c}^r \quad (1)$$

where $\mathbf{p}_i^{u'}$ is the aligned user joint, $\mathbf{c}^u$ and $\mathbf{c}^r$ are the user and reference centers, and s is the scale ratio. This alignment approach is robust to scale variation and partial occlusion. After alignment, the normalized user joint positions $\mathbf{p}_i^{u'}$ are used as inputs for subsequent joint angle computation and error estimation, ensuring that pose comparison is performed in a consistent coordinate space across users with different body proportions.

Trainer-Anchored Visual Guidance Module: This module reconstructs an implicit attention-guiding path based on the trainer's gaze direction extracted from the reference video. We estimate the gaze vector $\mathbf{g}_t$ at time t as the normal vector of the facial plane formed by Mediapipe facial landmarks—specifically the left eye, right eye, and nose tip:

$$\mathbf{g}_t = \text{normalize}((\mathbf{p}_{\text{RE}} - \mathbf{p}_{\text{LE}}) \times (\mathbf{p}_{\text{Nose}} - \mathbf{p}_{\text{LE}})) \quad (2)$$

where $\mathbf{p}_{\text{LE}}$, $\mathbf{p}_{\text{RE}}$, and $\mathbf{p}_{\text{Nose}}$ are the 3D positions of the left eye, right eye, and nose.

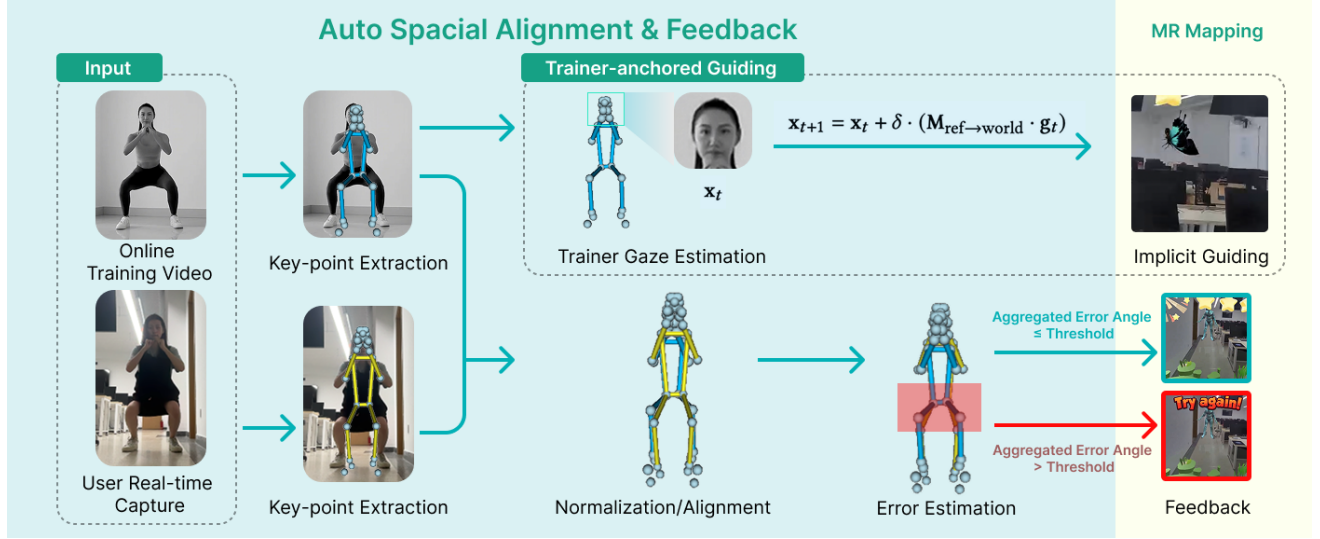


Figure 2: Overview of the Trainer-Anchored MR system for guided self-correction in postural rehabilitation.

This estimated gaze direction is transformed into the MR world coordinate system via a calibration matrix $M_{ref \rightarrow world}$, and used to generate a continuous spatial trajectory. Let x_t denote the 3D position of the visual guidance anchor in the MR world coordinate system at time step t .

$$x_{t+1} = x_t + \delta \cdot (M_{ref \rightarrow world} \cdot g_t) \quad (3)$$

The resulting trajectory is instantiated as a continuously moving visual guidance element (e.g., a butterfly) rendered in the MR environment. The position and motion of this element indicate the trainer's attention direction, guiding the user's visual focus toward relevant body parts or motion paths during exercise execution.

Pose Correction and Real-Time Feedback Module: This module compares key joint angles between the user and the reference. For joints forming a kinematic chain (a, b, c), the joint angle θ is computed as:

$$\theta = \cos^{-1} \left(\frac{(a-b) \cdot (c-b)}{\|a-b\| \cdot \|c-b\|} \right) \quad (4)$$

For each monitored joint, the system calculates the angular deviation $\Delta\theta_t$ between the user's current pose and the reference pose in real time, and compares it with a preset tolerance threshold. When the deviation falls within the permissible range, the system activates a dynamic "charging" visual feedback to encourage the user to maintain the correct posture. Once the deviation exceeds the threshold, the charging state is reset and accompanied by a "try again" voice prompt, while visual highlights or color warnings are triggered at the corresponding joint to clearly indicate the postural anomaly. Based on the value of $\Delta\theta_t$, the system performs threshold-based grading and color-coded visualization. Feedback is delivered through visual markers, thereby completing a closed-loop feedback process:

Perception → Comparison → Feedback → Correction

These visual cues dynamically adjust the properties of guiding elements (such as color, spatial offset, or animation intensity), prompting the user to adjust their posture in real time until the deviation falls within an acceptable range. Together, these modules form a robust and immersive training framework that facilitates natural and corrective movement learning through real-time MR interaction.

3 INTERACTIVE MECHANISMS AND SYSTEM DESIGN

Interaction feedback in our system is directly driven by pose error estimation, where different error patterns activate corresponding visual responses in the MR environment. To systematically design these responses, we categorize interaction mechanisms into two paradigms: point-based interactions and line-based interactions.

Point-based interactions operate on frame-fixed pose conditions and are suitable for static maintenance or critical pose attainment. When joint angles satisfy predefined spatial thresholds, the system triggers discrete visual feedback to indicate correctness, or accumulative feedback when the posture is maintained over a duration, reinforcing sustained performance.

Line-based interactions target dynamic movements characterized by continuous path changes. By monitoring joint motion along a kinematic trajectory, the system provides sequential or multi-point feedback that guides users through movement transitions. These interactions are designed to support complex exercises requiring coordinated motion rather than single-pose accuracy.

Building on these mechanisms, we implement a trainer-anchored self-correction training system that integrates real-time visual guidance and feedback within an MR environment, enabling users to iteratively adjust their posture during movement execution.

4 LIMITATIONS AND FUTURE WORK

Visual jitter may occur under large or fast movements, and the effectiveness of gaze-based guidance can be affected by head pose estimation and calibration accuracy, which we leave for future refinement and evaluation.

ACKNOWLEDGMENTS

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